Fields and Waves

Exam 2 Solutions

Fall 2025

1	**/12**
2	**/8**
3	**/12**
4	**/10**
Total	**/42**

Name _____

RIN _____

Notes:
SHOW ALL WORK. BEGIN WITH FORMULAS, THEN SUBSTITUTE VALUES AND UNITS. No credit will be given for numbers that appear without justification. Partial credit will be awarded where possible, so show your work so that we can follow your thought process. Use the backs of pages if there is not enough room on the front. If you do, please indicate that there is more work on the back of the page.

Unless otherwise stated in a problem, provide at least 2 significant digits in answers. Ensure that your calculator is set to use radians for trigonometric functions (and not degrees) and include units with all numerical answers.

At least skim through the entire quiz before you begin and then start with the problems you know best.

Problem 1: Gauss's Law with Spherical Symmetry [12 pts]

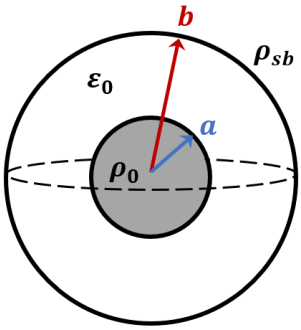
In this problem, you will use Gauss's law to solve for the electric field in each region.

The system below consists of:

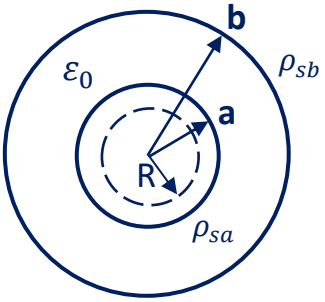
- A sphere of constant volumetric charge density ρ_0 and radius a.
- A conducting shell of radius b and surface charge density ρ_{sb}.

The total charge on the conducting shell is equal in magnitude to the total charge contained in the sphere, but opposite in sign. The region $a < r < b$ contains no free charge ρ and the permittivity is ε_0.

Note: All diagrams must include the shape and position of the Gaussian surface used in your solution.



a) [1 pt] Draw the diagram you will use for finding the electric field in the region $0 < r < a$.



b) [3 pts] Find the electric field in the region $0 < r < a$.

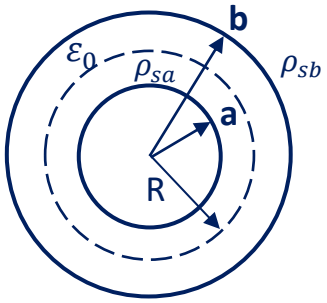
$$\oiint \vec{D} \cdot d\vec{s} = Q_{enc}$$

$$\oiint \vec{D} \cdot d\vec{s} = \int_0^{2\pi} \int_0^{\pi} \varepsilon_0 \vec{E}(R) \cdot R^2 \sin\theta \, \mathrm{d}\theta \mathrm{d}\phi = (2\pi)(2)\left(\varepsilon_0 R^2 E(R)\right) = 4\pi\varepsilon_0 R^2 E(R)$$

$$Q_{enc} = \int_0^{2\pi} \int_0^{\pi} \int_0^{R} \rho_0 r^2 \sin\theta \, \mathrm{d}r\mathrm{d}\theta\mathrm{d}\phi = (2\pi)(2)\rho_0 \left(\frac{R^3}{3}\right) = \frac{4}{3}\pi R^3 \rho_0$$

Solving for the electric field: $\vec{\boldsymbol{E}}(\boldsymbol{r}) = \frac{\boldsymbol{\rho_0}}{\boldsymbol{3\varepsilon_0}} \boldsymbol{r}\hat{\boldsymbol{r}}$

c) [1 pt] Draw the diagram you will use for finding the electric field in the region $a \leq r \leq b$.



d) [3 pts] Find the electric field in the region $a \leq r \leq b$.

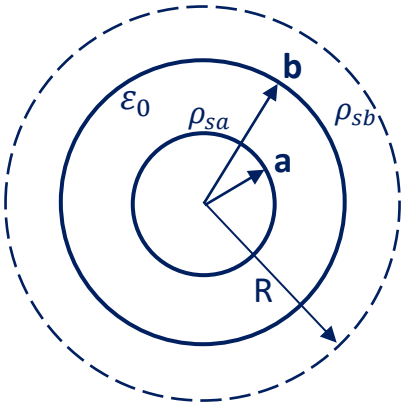
$$\oiint \vec{D} \cdot d\vec{s} = Q_{enc}$$

$$\int_0^{2\pi}\int_0^{\pi} \varepsilon_0 \vec{E}(R) \cdot R^2 \sin\theta \, \mathrm{d}\theta \mathrm{d}\phi = (2\pi)(2)\left(\varepsilon_0 R^2 E(R)\right) = 4\pi\varepsilon_0 R^2 E(R)$$

$$Q_{enc} = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} \rho_0 r^2 \sin\theta \, \mathrm{d}r\mathrm{d}\theta\mathrm{d}\phi = (2\pi)(2)\rho_0\left(\frac{a^3}{3}\right) = \frac{4}{3}\pi a^3 \rho_0$$

Solving for the electric field: $\boldsymbol{\vec{E}(r) = \frac{\rho_0}{3\varepsilon_0}\left(\frac{a^3}{r^2}\right)\hat{r}}$

e) [1 pt] Draw the diagram you will use for finding the electric field in the region $r > b$.



f) [1 pt] Find the electric field in the region $r > b$.

$$\oiint \vec{D} \cdot d\vec{s} = Q_{enc}$$

Since $Q_{enc} = Q_{inner} + Q_{outer} = 0$, $\boldsymbol{\vec{E}(\vec{r}) = 0 \text{ for } r > b}$.

g) [2 pts] If you were to replace the material in the region $a \leq r \leq b$ by a dielectric with $\varepsilon > \varepsilon_0$, how would this affect the potential difference between $r = b$ and $r = a$? Explain this change in the context of electric materials and polarization.

Placing a dielectric in this region would reduce the potential difference between the two locations, since the electric field in this region would be lower due to polarization - the formation of dipoles in the

dielectric would cause a response field to form that opposes the direction of the applied field and reduces the total field magnitude in the material.

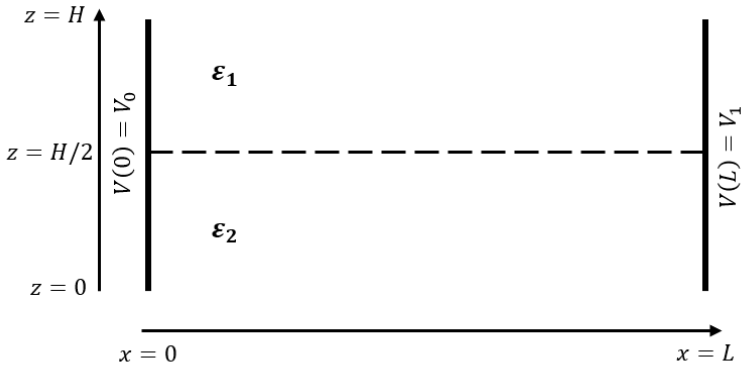
Problem 2: Parallel Plate Capacitor with Two Dielectrics [8 pts]

In the system below, the space between two conducting plates located at $x = 0$ and $x = L$ is filled with two different dielectrics:

- Region 1 occupies the space where $\frac{H}{2} < z \leq H$ and has permittivity ε_1.
- Region occupies the space where $0 \leq z < H/2$ and has permittivity ε_2.

The plate at $x = 0$ is held at a constant potential V_0, while the plate at $x = L$ is held at a constant potential V_1. The plates have height H in the z-dimension and depth W in the y-direction.

The electric field in region 1 is given by $\vec{E}_1(x) = \frac{V_1 - V_0}{L}\hat{x}$.



a) [2 pts] Find the electric field, $\vec{E}_2(x)$, and displacement field, $\vec{D}_2(x)$, in region 2. Ignore fringing effects.

Since the dielectric-dielectric interface is parallel to the x-y plane, the electric field in region 1 is tangent to the interface. Therefore $\vec{E}_1 = \vec{E}_{1t} = \vec{E}_{2t} = \boldsymbol{\vec{E}_2(x) = \frac{V_1 - V_0}{L}\hat{x}}$

The displacement field is given by $\boldsymbol{\vec{D}_2 = \varepsilon_2 \vec{E}_2(x) = \varepsilon_2 \frac{V_1 - V_0}{L}\hat{x}}$

b) [4 pts] Using your results from part a), find the total electric energy stored in the region between the conducting plates.

The total energy stored between the plates is given by $W_e = \frac{1}{2}\iiint (\vec{D}_1 \cdot \vec{E}_1) dV_1 + \frac{1}{2}\iiint (\vec{D}_2 \cdot \vec{E}_2) dV_2$

For region 1: $W_{e1} = \frac{1}{2}\int_{\frac{H}{2}}^{H}\int_0^W\int_0^L \left(\varepsilon_1 \frac{V_1 - V_0}{L}\right) \cdot \left(\frac{V_1 - V_0}{L}\right) dx dy dz = \frac{1}{4}\varepsilon_1 \left(\frac{V_1 - V_0}{L}\right)^2 HWL$

For region 2: $W_{e2} = \frac{1}{2}\int_0^{\frac{H}{2}}\int_0^W\int_0^L \left(\varepsilon_2 \frac{V_1 - V_0}{L}\right) \cdot \left(\frac{V_1 - V_0}{L}\right) dx dy dz = \frac{1}{4}\varepsilon_2 \left(\frac{V_1 - V_0}{L}\right)^2 HWL$

Total stored energy: $\boldsymbol{W_e = \frac{1}{4}\left(\frac{V_1 - V_0}{L}\right)^2 (\varepsilon_1 + \varepsilon_2) HWL}$

c) [2 pts] Find the capacitance of the system using the total stored electric energy method.

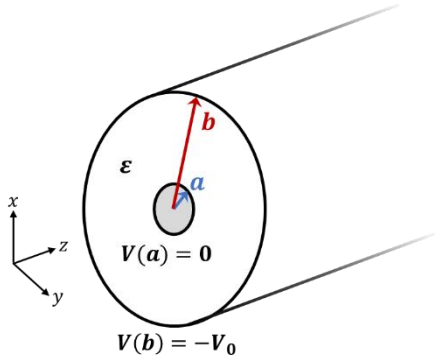
$$W_e = \frac{1}{2}CV^2 \rightarrow C = \frac{2W_e}{V^2} = \frac{2\left(\frac{1}{4}\left(\frac{V_1 - V_0}{L}\right)^2 (\varepsilon_1 + \varepsilon_2)HWL\right)}{(V_1 - V_0)^2} = (\boldsymbol{\varepsilon_1 + \varepsilon_2})\frac{\boldsymbol{HW}}{\boldsymbol{2L}} = \varepsilon_1\frac{HW}{2L} + \varepsilon_2\frac{HW}{2L}$$

d) [2 pts] **Bonus**: Does your result in part c) agree with how parallel capacitances combine according to circuit theory? Explain.

Yes, capacitances in parallel sum directly and the capacitance of this system is the sum of the capacitance of region 1 plus the capacitance of region 2.

Problem 3: Coaxial Cable – Laplace's Equation [12 pts]

The coaxial cable below consists of a grounded, inner conductor of radius a and an outer conducting shell of radius b, which is held at a constant potential $-V_0$. The space between the conductors $a < r < b$ has no free charge and is filled by a dielectric with permittivity ε.



a) [6 pts] Use Laplace's or Poisson's equation to find $V(\vec{r})$ in the region $a \le r \le b$.

$$\nabla^2 V(\vec{r}) = 0$$

General solution:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) = 0 \rightarrow r\frac{\partial V}{\partial r} = C_0 \rightarrow \boldsymbol{V(r) = C_0 \ln r + C_1}$$

Boundary conditions:

(1) $V_1(r = a) = \boldsymbol{0 = C_0 \ln a + C_1}$

(2) $V_2(r = b) = \boldsymbol{-V_0 = C_0 \ln b + C_1}$

(1) – (2): $V_0 = C_0 \ln\left(\frac{a}{b}\right) \rightarrow C_0 = \frac{V_0}{\ln\left(\frac{a}{b}\right)}$

Plugging into (1): $0 = \frac{V_0}{\ln\left(\frac{a}{b}\right)} \ln a + C_1 \rightarrow C_1 = -\frac{V_0}{\ln\left(\frac{a}{b}\right)} \ln a$

Particular solution:

$$V(r) = \frac{V_0}{\ln\left(\frac{a}{b}\right)} \ln r - \frac{V_0}{\ln\left(\frac{a}{b}\right)} \ln a = \frac{\boldsymbol{-V_0}}{\boldsymbol{\ln\left(\frac{b}{a}\right)}} \boldsymbol{\ln\left(\frac{r}{a}\right)}$$

b) [2 pts] Find the displacement field $\vec{D}(\vec{r})$ in the region $a \le r \le b$.

$$\vec{D}(r) = \varepsilon\vec{E}(r) = -\varepsilon\nabla V(r) = -\frac{d}{dr}\left[\frac{-\varepsilon V_0}{\ln\left(\frac{b}{a}\right)} \ln\left(\frac{r}{a}\right)\right] = \frac{\varepsilon V_0}{\ln\left(\frac{b}{a}\right)}\left(\frac{a}{r}\right)\left(\frac{1}{a}\right)\hat{r} = \boldsymbol{\frac{\varepsilon V_0}{\ln\left(\frac{b}{a}\right)}\left(\frac{1}{r}\right)\hat{r}}$$

c) [2 pts] Find the charge density on the inner and outer conductors.

For the inner conductor: $D_{1n} - D_{2n} = \rho_{sa} \rightarrow \rho_{sa} = D_{1n} = \boldsymbol{\frac{\varepsilon V_0}{\ln\left(\frac{b}{a}\right)}\left(\frac{1}{a}\right)}$

For the outer conductor: $D_{1n} - D_{2n} = \rho_{sb} \rightarrow \rho_{sb} = -D_{2n} = -\frac{\boldsymbol{\varepsilon V_0}}{\boldsymbol{\ln\left(\frac{b}{a}\right)}}\left(\frac{\boldsymbol{1}}{\boldsymbol{b}}\right)$

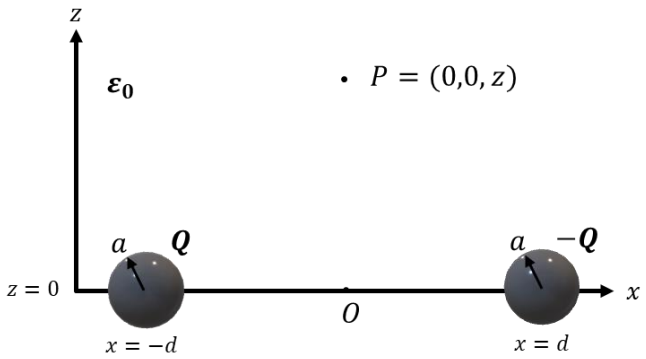
d) [2 pts] Find the capacitance per unit length of the coaxial cable.

$C = \frac{Q}{V} = \frac{A\rho_{sa}}{V_0 L} = \frac{2\pi a \rho_{sa}}{V_0} = \frac{2\pi a}{V_0}\frac{\varepsilon V_0}{\ln\left(\frac{b}{a}\right)}\left(\frac{1}{a}\right) = \boldsymbol{\frac{2\pi\varepsilon}{\ln\left(\frac{b}{a}\right)}}$

Problem 4: Superposition of Electric Fields [10 pts]

Two conducting spheres of radius a are located on the x-axis: One sphere is located at $x = -d$ and has total surface charge Q, while the other sphere is located $x = d$ and has total surface charge $-Q$.

A diagram of this system is shown below.



a) [2 pts] First, consider a conducting sphere of radius a and total surface charge Q located at the origin. Find the electric field due to this sphere for $r > a$ (outside the sphere).

Using Gauss's law $\iint \vec{D} \cdot d\vec{s} = Q_{enc}$ and a Gaussian sphere of radius $R > a$:

$\iint \vec{D} \cdot d\vec{s} = \int_0^{2\pi} \int_0^{\pi} E(R)\hat{r} \cdot R^2 \sin\theta \, d\theta d\phi \hat{r} = 4\pi E(R) R^2$ and $Q_{enc} = Q$

Solving for the electric field: $\boldsymbol{\vec{E}(r) = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{r}}$

b) [4 pts] Find the total electric field due to both of the conducting spheres in the figure above at the observation point $P = (0,0,z)$. You must express your answer in terms of cartesian coordinates.

Vector pointing from origin to observation point P: $\vec{r} =< 0,0,z >$

Electric field of sphere with charge Q

Vector pointing from origin to sphere with charge Q: $\vec{r}_{0+} =< -d,0,0 >$

Vector pointing from center of sphere with charge Q to P: $\vec{r}_+{}' =< d,0,z >$

$$\vec{E}_+(r) = \frac{Q}{4\pi\varepsilon_0 |\vec{r}_+{}'|^2}\hat{r}'_+ = \frac{Q}{4\pi\varepsilon_0 |\vec{r}_+{}'|^3}\vec{r}'_+ = \boldsymbol{\frac{Q}{4\pi\varepsilon_0 (d^2+z^2)^{\frac{3}{2}}} < d,0,z >}$$

Electric field of sphere with charge -Q

Vector pointing from origin to sphere with charge -Q: $\vec{r}_{0-} =< d,0,0 >$

Vector pointing from center of sphere with charge -Q to P: $\vec{r}'_- =< -d,0,z >$

$$\vec{E}_-(r) = \frac{-Q}{4\pi\varepsilon_0 |\vec{r}_-{}'|^2}\hat{r}'_- = \frac{-Q}{4\pi\varepsilon_0 |\vec{r}_-{}'|^3}\vec{r}'_- = \boldsymbol{\frac{Q}{4\pi\varepsilon_0 (d^2+z^2)^{\frac{3}{2}}} < d,0,-z >}$$

Total electric field due to both spheres

$$\vec{\boldsymbol{E}}_{\boldsymbol{tot}}(\boldsymbol{r}) = \vec{\boldsymbol{E}}_{+}(\boldsymbol{r}) + \vec{\boldsymbol{E}}_{-}(\boldsymbol{r}) = \frac{\boldsymbol{Qd}}{\boldsymbol{2\pi\varepsilon_0(d^2+z^2)^{\frac{3}{2}}}}\hat{\boldsymbol{x}}$$

c) [1 pt] A point charge with charge $-Q$ is now placed at observation point P. What is the force on the point charge due to the two conducting spheres?

$$\vec{F}(r) = q\vec{E}(r) = -Q * \frac{Qd}{2\pi\varepsilon_0(d^2+z^2)^{\frac{3}{2}}}\hat{x} = -\frac{\boldsymbol{Q^2d}}{\boldsymbol{2\pi\varepsilon_0(d^2+z^2)^{\frac{3}{2}}}}\hat{\boldsymbol{x}}$$

d) [2 pts] Set up an integral for calculating the potential difference between the origin $O = (0,0,0)$ and the observation point $P = (0,0,z)$. You do not need to solve it!

$$\boldsymbol{V(P) - V(O) = -\int_0^z \frac{Qd}{2\pi\varepsilon_0(d^2+z^2)^{\frac{3}{2}}}\hat{x}\cdot dz\hat{z} = 0}$$

e) [1 pt] In electrostatics, does the path of integration matter when calculating potential differences? Which of Maxwell's equations for electrostatics tells us this?

No, the path does not matter. $\oint \vec{E}\cdot d\vec{l} = 0$ tells us that we can take any path we want, but only the evaluation of the integral at the end points matters.